

Project Guidelines

I. Overall guideline:

The project should apply data science methods to answer an interesting economics question. This echoes the soul of this course, which combines data science and economics.

Data science part: You don't have to use machine learning or even more complicated methodologies in the project. Sometimes, if you choose an interesting economics question, and you only use satellite and imagery data as data source, it's still within the overall framework of the project requirement. But it will be great if you can apply multiple methodologies (e.g. causal inference, machine learning) and data sources (e.g. text data, satellite and imagery data, other big data) to the final project.

Economics question part: The final topic for the project should be an economics question. An economics research question is a clear, focused problem or puzzle about how economic agents (individuals, households, firms), markets, institutions, or policies behave or interact, stated so it can be answered empirically or theoretically. This course is more about empirical, so you'll need to answer it empirically.

II. Some guidelines for choosing a good research question:

A good economics research question should typically have the following elements:

- Specific and answerable (not overly broad).
- Economically meaningful (addresses cause, mechanism, welfare, or policy).
- Feasible (data or methods exist to analyze it).
- Novel (adds new evidence, theory, or perspective).
- Testable (yields clear hypotheses and observable implications).

Let me give you several good examples, and hope you can have a better sense through them.

Example 1:

Netzer, O., Lemaire, A., & Herzenstein, M. (2019). When words sweat: Identifying signals for loan default in the text of loan applications. *Journal of Marketing Research*, 56(6), 960-980.

Why is this a good research question, especially for this course?

1. It answers an economically meaningful question: do we have better ways to improve

the efficiency of decreasing the loan default for lending institutions? If yes, what is it? How is economically meaningful of these new ways if they exist?

2. It applies data science to this question. The authors collect the the raw text in over 120,000 loan requests from a lending institution and apply text mining methods to find useful information within this text. Specifically, they have used:
 - a. TF-IDF
 - b. LDA model
 - c. Regularized logistic regressions
 - d. Tree-based methods (random forest and extra trees)
3. The paper finds meaningful results. Particularly, they find that using these text data can increase lenders' return on investment (ROI) over an approach that uses only financial and demographics information by 5.75%. This is a substantial impact, and can bring real money to the firms.
4. Novelty. This paper fills several gaps and brings new evidence in the literature. Particularly, it provides new evidence on using application raw text data as a source information to predict writers' distant future behavior. The paper has received 393 Google citations since its publication, which embodies its impact.

When words sweat: Identifying signals for loan default in the text of loan applications
O Netzer, A Lemaire, M Herzenstein
Journal of Marketing Research 56 (6), 960-980

343

2019

Example 2:

Taylor, C. A., & Druckenmiller, H. (2022). Wetlands, flooding, and the clean water act. *American Economic Review*, 112(4), 1334-1363.

Why is this a good research question, especially for this course?

1. It tries to deal with a challenging question and it is policy relevant (relates to the federal policy about what is the right definition of 'waters of the United States' (WOTUS).
2. It applies data science to this question. The authors collect rich satellite and GIS data to identify wetlands, waters, measure distances, and identify upstream and downstream. They also collect big data on insurance claims (over 2 million transactions). Specifically, they have used:
 - a. CRS
 - b. Buffer, distance, spatial join
 - c. Regressions and causal inference strategies (causal inference strategies are not required in this class, but it's definitely great to use)
3. Economically meaningful findings. For example, they estimate that each hectare of wetland loss increases residential flood damages by approximately 12,300. This will be good evidence to use in the policy debate.
4. This paper is also novel in the topic, the identification strategy, and the empirical findings. This leads to 120 Google citations in just four years.

III. Some useful tips for finding research questions:

1. Read other papers and news articles. For example, NBER working paper series is a good source. It is a frontier of top economics and finance research.
https://www.nber.org/papers?page=1&perPage=50&sortBy=public_date
2. Check the professors and the companies of your interest. The professors usually put their research on the website, including their publications, working papers, and even work in progress. Then you can sense some potential research questions. This can also work for some companies, where you can find some topics they are interested in, and potentially have some research overlap or even collaborations. For example, Prof. Nick Bloom at Stanford. <https://nbloom.people.stanford.edu/>
3. Your own observation from your daily life or experiences. But you have to move much forward to formulate a economically meaningful research question.

IV. The evaluation criteria:

1. Overall project quality (coherence, clarity, significance)
2. Research question (clarity, focus, testability)
3. Novelty and contribution (originality, relation to literature)
4. Rigor of data-science methods (appropriateness, implementation, validation)
5. Data quality and appropriateness (sources, cleaning, limitations)
6. Effectiveness of data visualization & presentation (clarity, accuracy, storytelling)
7. Usefulness / policy relevance (practical implications)
8. Reproducibility & code quality (documentation, ability to reproduce results)